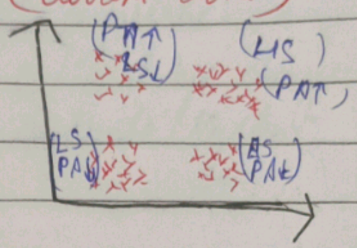
Unsupervised Learning

⟶ Here target variable is not given

⟶ → It groups/segments your data
(find patterns in data)
⇓
(current data)

eg → we are running a store

data: customer Salary, purchase Amount



Why UL → this grouping can be seen using a basic scatter plot.

⟶ in higher -D you can-not identify groups/patterns by seeing a plot.

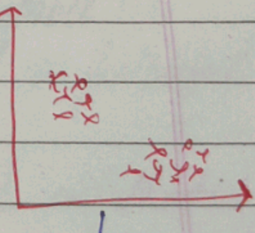
Types of UL

1. K means clustering
2. Hierarchical clustering
3. DB Scan

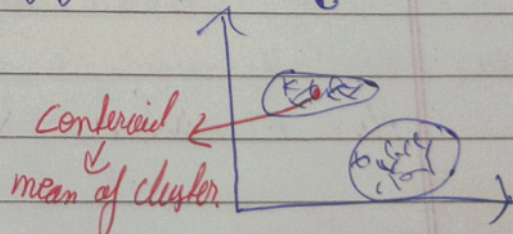
(1) K-means-clustering

means ⇒ mean

K ⇒ no. of groups



↓



Steps of K-mean- clustering

(1) initilize the centeraids (K)

↳ consider any random datapo[int] as centeraid.

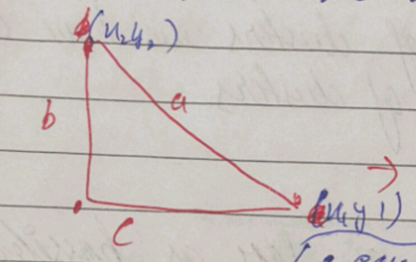
↳ you can take avg of all dat[a] as centeraid value

(2) Point near to centeraid will be labelled as the groups.

(3) Recalculate the centeraid. (Average).

Repeat step 2 & 3 until centeraid doesn't chage.

* How to calculate the distance



$\rightarrow \quad a^2 = c^2 + b^2$

$a = \sqrt{c^2 + b^2}$

- euclidean distance

$$a = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

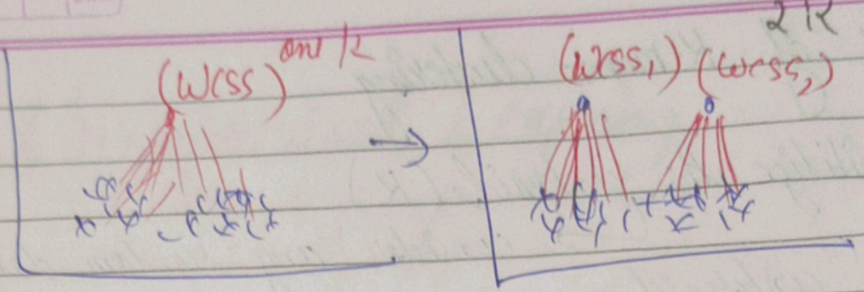
- manhattan distance

$$a = |x_2 - x_1| + |y_2 - y_1|$$

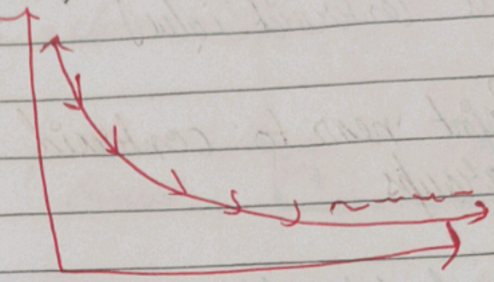
How to Select - K-values

WCSS → within cluster sum of squared distance

$$WCSS = \sum_{i=1}^{n} \left(\text{distance b/w points to nearest centeraid} \right)^2$$



* So as the K ↑, WCSS ↓



graph of WCSS
↓
The WCSS value keep decreasing as the no. of clusters K increases.
⇓
At one point the WCSS becomes almost constant
⇓
At that time the no of clusters we have will be the optimal no. of clusters

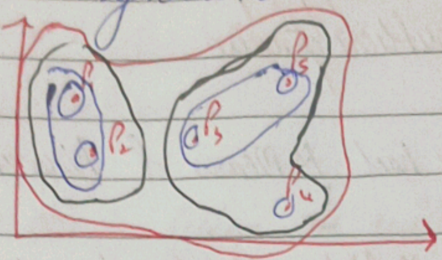
NOTE → Initilize K as far as possible

② Hirarchical Clustering

↳ in this clustering we don't need any k

It is of two types

① Aglomerative



steps →

① each data point is cluster in its own (no. of cluster = no. of data point)

② find the nearest cluster and combine to create a new cluster until we get a single cluster.

* How to perform steps?
  - using → • euclidean distance
    - manhattan distance
    - Cosine Similarity

② Divisive clustering

steps

① generate distance matrix

② Create minimum spanning tree based on distance b/w data point using prims/Kruskal's algo

③ Use Complete linkage to form cluster

④ Repeat step 3 until all individual data point are represents as individual clusters

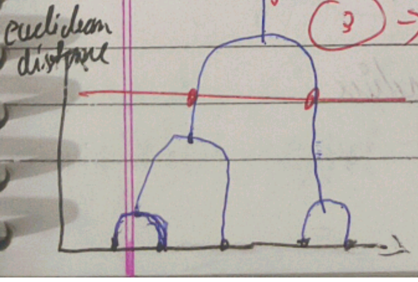
* How to select K?

↳ using dindograms → make a horizonal line → no. of point it cuts dindogram → is the no of k.

① Business team will help you out in determining k

② Using a thres hold → also business team will tell.

③ → Trick ⇒ we can take longest vertical line of dindogram where none of horizontal of dindogram is passing → draw horizontal line then no. of cuts = no of clusters.

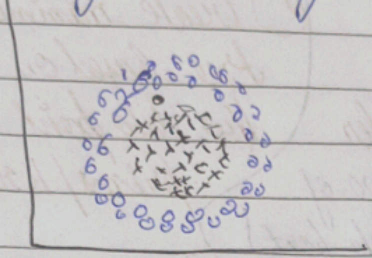


DBSCAN {Density Based Spatial Clustering of Application with Noise}

* characteristic of DBSCAN
  - used for finding Patterns/groups
  - used for finding outliers/noise

Que → Why DBSCAN if we had K-Means & hierarchical

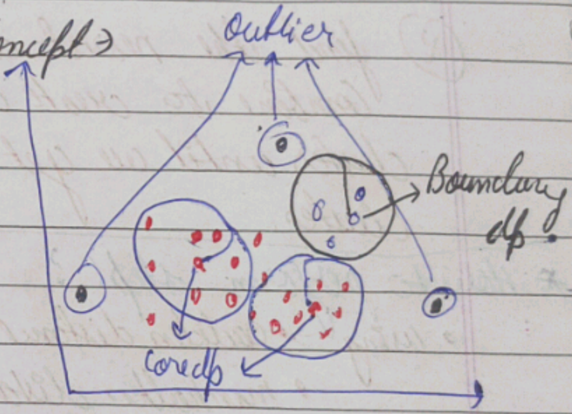
eg →



* Distance based algo such as K-means & hierarchical can not be used here.
* The data is non-linear.

it works in a simple concept →

1. Core Data points
2. Boundary datapoints
3. Outlier
4. Minimum no. of dp
5. Radius (ε)



(1) Core dp → The dp's which have atleast minimum number of dp's in its radius

(2) Boundary dp's → The dp's which donot have min no. of dp's within the radius → dp → (1 ≤ dp < min)

(3) Outlier → No dp in its radius

Silhouette Score / coefficient

for classification → accuracy
regression → r^2 square

→ Similarly silhouette score is metric for clustering algorithm.

* Que → When we will say given ~~at~~ cluster is good cluster?

① within cluster data points are compact.

tightly packed / very closed to each other

→ WCSS → it should be minimum

② Outside cluster sum of squares should be maximum as much as possible ~~fr~~ from nearest cluster.

→ OCSS → it should be maximum.

WCSS ⇒ a(i) = summation → distance within cluster w.r.t each dp.

OCSS → b(i) → distance from nearest cluster

$$\text{Silhouette Score } S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

→ range [-1 to 1]

$$a(i) = \frac{1}{|C_i| - 1} \sum_{j \in C_i, j \neq i} d(ij)$$

$$b(i) = \min_{k \neq i\ \text{cluster}} d(C_k, i)$$